

# CV: John Nicholson

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| EMPLOYMENT   | <b>Rankin-Sneddon Research Fellow, University of Glasgow</b><br>Promoted to Grade 8 (August 2025)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Sep 2023 - Sep 2026  |
|              | <b>Heilbronn Research Fellow, Imperial College London</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Oct 2021 - Sep 2023  |
| EDUCATION    | <b>University College London</b><br>PhD Mathematics, EPSRC PhD Studentship (EP/N509577/1)<br>Thesis: Projective modules over integral group rings and Wall's D2 problem<br>Supervisor: Francis E. A. Johnson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Oct 2017 - Oct 2021  |
|              | <b>University of Cambridge, Emmanuel College</b><br>MMath Mathematics (Part III), First Class Honours<br>MA Mathematics, First Class Honours in all years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Sep 2013 - July 2017 |
| PUBLICATIONS | <ol style="list-style-type: none"><li>12. <b>Exotic presentations of quaternion groups and Wall's D2 problem</b><br/>with Tommy Hofmann, 36 pages, <a href="https://arxiv.org/abs/2507.15999">arXiv:2507.15999</a> (2025)<br/>Journal of Algebra, to appear</li><li>11. <b>Simple homotopy types of even dimensional manifolds</b><br/>with Csaba Nagy and Mark Powell, 57 pages, <a href="https://arxiv.org/abs/2312.00322">arXiv:2312.00322</a> (2023)<br/>Annales Scientifiques de l'École Normale Supérieure, to appear</li><li>10. <b>Four-manifolds, two-complexes and the quadratic bias invariant</b><br/>with Ian Hambleton<br/>Advances in Mathematics, 486 (2026), 110728, 69 pages</li><li>9. <b>Stably free modules and the unstable classification of 2-complexes</b><br/>Mathematische Annalen, 394 (2026), 66, 41 pages</li><li>8. <b>Stable equivalence relations on 4-manifolds</b><br/>with Daniel Kasprowski and Simona Veselá<br/>Proceedings of the London Mathematical Society, 131 (2025), e70101, 37 pages</li><li>7. <b>On the universal pairing for 2-complexes</b><br/>with Mikhail Khovanov and Slava Krushkal<br/>Bulletin of the London Mathematical Society, 57 (2025), 2838-2853</li><li>6. <b>Cancellation for <math>(G, n)</math>-complexes and the Swan finiteness obstruction</b><br/>International Mathematics Research Notices, 2024 (2024), 11832-11864</li><li>5. <b>Projective modules and the homotopy classification of <math>(G, n)</math>-complexes</b><br/>Algebraic &amp; Geometric Topology, 24 (2024), 2245-2284</li><li>4. <b>Homotopy classification of 4-manifolds whose fundamental group is dihedral</b><br/>with Daniel Kasprowski and Benjamin Ruppik<br/>Algebraic &amp; Geometric Topology, 22 (2022), 2915-2949</li><li>3. <b>Connected sum decompositions of high-dimensional manifolds</b><br/>with I. Bokor, D. Crowley, S. Friedl, F. Hebestreit, D. Kasprowski and M. Land<br/>2019-20 MATRIX Annals (2021), 5-30</li><li>2. <b>On CW-complexes over groups with periodic cohomology</b><br/>Transactions of the American Mathematical Society, 374 (2021), 6531-6557</li></ol> |                      |

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|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                         | 1. <b>A cancellation theorem for modules over integral group rings</b><br>Mathematical Proceedings of the Cambridge Philosophical Society, 171 (2021), 317-327                                                                   |             |
| PREPRINTS               | 16. <b>A survey on mapping class groups of 3-manifolds</b><br>with P. Bader, R. Boyd, G. Carfora, G. Corrigan, D. Galvin, C. Nagy, W. Niu, I. Nonino, M. Pencovitch and M. Powell, 82 pages (2026)                               |             |
|                         | 15. <b>Swan modules and homotopy types after a single stabilisation</b><br>with Tommy Hofmann, 19 pages, <a href="#">arXiv:2507.21975</a> (2025)                                                                                 |             |
|                         | 14. <b>Generalised doubles and simple homotopy types of high dimensional manifolds</b><br>with Csaba Nagy and Mark Powell, 25 pages, <a href="#">arXiv:2409.03082</a> (2024)                                                     |             |
|                         | 13. <b>The cancellation property for projective modules over integral group rings</b><br>39 pages, <a href="#">arXiv:2406.08692</a> (2024)                                                                                       |             |
| ARTICLES IN PREPARATION | 21. <b>Simple homotopy types of four-manifolds and generalised <math>\mathcal{N}</math>-torsion</b><br>with Csaba Nagy. Draft available upon request                                                                             |             |
|                         | 20. <b>The Swan finiteness obstruction for hyperelementary periodic groups</b><br>with James F. Davis. Draft available upon request                                                                                              |             |
|                         | 19. <b>Surfaces in 4-manifolds: a compendium</b><br>with V. Bais et al. (44 coauthors). Draft available upon request                                                                                                             |             |
|                         | 18. <b>Simple homotopy types of manifolds with abelian fundamental groups</b><br>Draft available upon request                                                                                                                    |             |
|                         | 17. <b>Exotic presentations and relation modules of 2-dimensional groups</b><br>with Grigori Avramidi and Max Forester. Draft available upon request                                                                             |             |
| FUNDING AND AWARDS      | International Centre for Mathematical Sciences, Workshop Award. PI. £14,000.                                                                                                                                                     | 2025        |
|                         | Research Workshop Grant (Scheme 1), London Mathematical Society. PI. £5,500.                                                                                                                                                     | 2025        |
|                         | Small Grants Scheme, Heilbronn Institute (x4). Four grants. PI. Total £9,400.                                                                                                                                                    | 2024, 2025  |
|                         | Research Support Fund, Glasgow Mathematical Journal Trust. Co-PI. £1,500.                                                                                                                                                        | 2025        |
|                         | Research Support Fund, Edinburgh Mathematical Society. Co-PI. £1,500.                                                                                                                                                            | 2025        |
|                         | EPSRC Vacation Bursary Award (x4). To support four projects. Total £15,012.                                                                                                                                                      | 2023, 2024  |
|                         | John Robertson Bequest, University of Glasgow. PI. £1,250.                                                                                                                                                                       | 2024        |
|                         | Rankin-Sneddon Research Fellowship, University of Glasgow                                                                                                                                                                        | 2023 - 2026 |
|                         | Focused Research Grant, Heilbronn Institute. PI. £8,000.                                                                                                                                                                         | 2023        |
|                         | Cecilia Tanner Research Impulse Grant, Imperial College London. PI. £800.                                                                                                                                                        | 2023        |
|                         | Fields Research Fellowship. To visit Fields Institute, Canada. £4900.                                                                                                                                                            | 2023        |
|                         | Heilbronn Research Fellowship, Heilbronn Institute £189,852.                                                                                                                                                                     | 2021 - 2023 |
|                         | Guest Program, Max Planck Institute for Mathematics. 2-month visit, £2,600.                                                                                                                                                      | 2021        |
|                         | EPSRC PhD DTP Studentship. 4-year studentship. Salary, Fees & £1,400 p.a.                                                                                                                                                        | 2017 - 2021 |
|                         | Undergraduate Awards: Braithwaite Batty Prize for Exceptional Exam Results in Mathematics (x2), Senior Scholar (x3), Rowley Mainhood Award (x2), Emmanuel College Prize (x2), Bridgewater Studentship for Undergraduate Research | 2014 - 2017 |
| SELECTED INVITED TALKS  | Cohomology of Finite Groups: Interactions and Applications, Oberwolfach                                                                                                                                                          | Mar 2026    |
|                         | Scottish Topology Seminar, ICMS, Edinburgh                                                                                                                                                                                       | Oct 2025    |
|                         | Topological 4-manifolds, CRM, University of Montreal                                                                                                                                                                             | July 2025   |
|                         | Functor Categories for Groups, Lancaster University                                                                                                                                                                              | Apr 2025    |
|                         | Boise State University, Algebra and Topology Seminar                                                                                                                                                                             | Apr 2025    |
|                         | University of Essex, Mathematics Essex Seminar Series                                                                                                                                                                            | Dec 2024    |
|                         | Universidad Autónoma de Madrid (ICMAT), Group Theory Seminar                                                                                                                                                                     | Oct 2024    |
|                         | Max Planck Institute for Mathematics, Physical Math Seminar                                                                                                                                                                      | Sep 2024    |
|                         | Focused Research Group on 4-Manifolds, University of Glasgow                                                                                                                                                                     | June 2024   |
|                         | Transalpine Topology Tetrahedron, University of Warwick                                                                                                                                                                          | Apr 2024    |

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|                                  | University of Cambridge, Differential Geometry & Topology Seminar              | Feb 2024    |
|                                  | University of Glasgow, Algebra & Number Theory Seminar                         | Jan 2024    |
|                                  | University of Glasgow, Geometry & Topology Seminar                             | Nov 2023    |
|                                  | University of Augsburg, Differential Geometry Seminar                          | July 2023   |
|                                  | University of Southampton, Topology Seminar                                    | May 2023    |
|                                  | McMaster University, Geometry & Topology Seminar                               | Apr 2023    |
|                                  | University of Toronto, Geometry & Topology Seminar                             | Apr 2023    |
|                                  | The London Algebra Colloquium                                                  | Feb 2023    |
|                                  | University of Exeter, Number Theory and Algebra Seminar                        | Dec 2022    |
|                                  | Geometric Group Theory and Low Dimensional Topology, 3rd International Meeting | Nov 2022    |
|                                  | University of Oxford, Topology Seminar                                         | Oct 2022    |
|                                  | Early Career Researchers in Mathematics 2022, University College London        | June 2022   |
|                                  | University of Münster, Geometric Group Theory Seminar                          | June 2022   |
|                                  | Indiana University Bloomington, Topology Seminar                               | Apr 2022    |
|                                  | Special Session on Geometric Topology, AMS Spring Sectional Meeting            | Mar 2022    |
|                                  | The London Geometry and Topology Seminar                                       | Jan 2022    |
|                                  | The University of Alabama, Algebra/Topology Seminar                            | Nov 2021    |
|                                  | Universidad de Buenos Aires, Topology Seminar                                  | Oct 2021    |
|                                  | LMU München, Arithmetic & Algebraic Geometry Seminar                           | June 2021   |
|                                  | Université du Québec à Montréal, Geometry & Topology Seminar                   | Apr 2021    |
|                                  | Max Planck Institute for Mathematics, Topology Seminar                         | Mar 2021    |
|                                  | New Directions in Group Theory and Triangulated Categories                     | Dec 2020    |
|                                  | University of Southampton, Pure Mathematics Colloquium                         | Nov 2020    |
|                                  | University of Regensburg, Low-Dimensional Topology Seminar                     | Oct 2020    |
|                                  | Ohio State University, Topology Seminar                                        | July 2020   |
|                                  | Durham University, Geometry and Topology Seminar                               | Oct 2019    |
|                                  | University of Melbourne, Topology Seminar                                      | Aug 2018    |
|                                  | University of Sydney, Geometry and Topology Seminar                            | Aug 2018    |
| SELECTED<br>CONTRIBUTED<br>TALKS | 37th British Topology Meeting, University of Aberdeen                          | Aug 2024    |
|                                  | Topology, representation theory and higher structures, Isle of Skye, Scotland  | May 2024    |
|                                  | Topological and Homological Methods in Group Theory, Bielefeld University      | Mar 2022    |
|                                  | Four Manifolds: Confluence of High and Low Dimensions, Fields Institute        | July 2019   |
| UNIVERSITY<br>TEACHING           | <b>School of Mathematics &amp; Statistics, University of Glasgow</b>           |             |
|                                  | <i>Lecturer:</i> Algebraic and Geometric Topology (35 lectures, 35 students)   | 2025 - 2026 |
|                                  | <i>Lecturer:</i> Galois Theory (18 lectures, 13 students)                      | 2024 - 2025 |
|                                  | <i>Lecturer:</i> Linear Algebra (21 lectures, 240 students)                    | 2024 - 2025 |
|                                  | <i>Lecturer:</i> Galois Theory (18 lectures, 30 students)                      | 2023 - 2024 |
|                                  | <i>Lecturer:</i> Linear Algebra (21 lectures, 70 students)                     | 2023 - 2024 |
|                                  | <b>Department of Mathematics, Imperial College London</b>                      |             |
|                                  | <i>Lecturer:</i> Algebra III (24 lectures, 50 students)                        | 2022 - 2023 |
|                                  | <b>Department of Mathematics, University College London</b>                    |             |
|                                  | <i>Problems class instructor:</i> Algebra III (2 classes, 30 students each)    | 2019 - 2020 |
| STUDENT<br>SUPERVISION           | <i>Problems class instructor:</i> Algebra IV (130 Students)                    | 2018 - 2019 |
|                                  | <i>Tutor:</i> Algebra I, Algebra II, Analysis I, Analysis II (7 Students)      | 2018 - 2019 |
|                                  | <i>Problems class assistant:</i> Algebra II (70 students)                      | 2017 - 2018 |
|                                  | <b>Undergraduate Research Project, University of Glasgow</b>                   | 2023-24     |
|                                  | <i>Iain McWhinnie</i> (Title: Exotic presentations for the Klein bottle group) |             |

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|                                   | <b>Undergraduate Research Project, University of Glasgow</b><br>(supervised jointly with Philipp Bader, PhD student at University of Glasgow)<br><i>Ryan Lamy</i> (Title: Research in surface theory)<br>Resulted in a research article “Roots of hyperelliptic involutions and braid groups modulo their center inside mapping class groups” (see <a href="https://arxiv.org/abs/2503.09552">arXiv:2503.09552</a> )<br>Currently a PhD student at Université de Caen Normandie (2025 - )                                                                                                                                                    | 2023-24                                          |
|                                   | <b>Undergraduate Research Project, Imperial College London</b><br><i>William Thomas</i> (Title: Geometric realisation over aspherical groups)<br>Winner of the prize for Best Undergraduate Research Project in Mathematics<br>Resulted in a research article “Geometric realisation of relation modules over aspherical groups” published in <i>Topology and its Applications</i> (see <a href="https://arxiv.org/abs/2312.02948">arXiv:2312.02948</a> )<br>Currently a PhD student at the University of Oxford (2024 - )                                                                                                                   | 2022-23                                          |
|                                   | <b>Undergraduate Research Project, Imperial College London</b><br><i>Lukas Ertl</i> (Title: Relation modules over Baumslag-Solitar groups)<br>Currently a PhD student at Swansea University (2024 - )                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2022-23                                          |
|                                   | <b>MSc Thesis, Imperial College London</b><br><i>Kamen Pavlov</i> (Title: The Wall finiteness obstruction)<br>Winner of the prize for Best MSc Project in Pure Mathematics<br>Currently a PhD student at the University of Southampton (2023 - )                                                                                                                                                                                                                                                                                                                                                                                             | 2022-23                                          |
|                                   | <b>MSc Thesis, Imperial College London</b> (supervised jointly with Travis Schedler)<br><i>Alexandre Tissot</i> (Title: On group cohomology and Poincaré duality groups)                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2021-22                                          |
| ACADEMIC SERVICE                  | <b>Peer Review and Quick Opinions</b> <i>Mathematische Annalen</i> , <i>Proceedings of the London Mathematical Society</i> , <i>Journal of Topology</i> , <i>Journal of the London Mathematical Society</i> , <i>Algebraic &amp; Geometric Topology</i> , <i>K3 Project: a new problem list in low-dimensional topology</i> , <i>Journal of Pure and Applied Algebra</i> , <i>Homology Homotopy and Applications</i> , <i>Israel Journal of Mathematics</i> , <i>Beiträge zur Algebra und Geometrie</i> .<br><br><b>Grant Review</b> EPSRC Postdoctoral Research Fellowship<br><br><b>EPSRC Pure Mathematics Community Workshop</b> Attendee | Mar 2023                                         |
| WORKSHOP AND SEMINAR ORGANISATION | <b>Geometry &amp; Topology Seminar, University of Glasgow</b><br>Sole organiser: 15 external speakers so far (5 UK, 10 international)<br>Sole organiser: 21 external speakers (9 UK, 12 international)<br>Organiser (with Jim Belk): 20 external speakers (8 UK, 12 international)                                                                                                                                                                                                                                                                                                                                                           | 2025 - 2026<br>2024 - 2025<br>2023 - 2024        |
|                                   | <b>Research Workshop: “Mapping class groups of non-simply connected 4-manifolds”</b><br>Organiser (with Michelle Daher and Mark Powell)<br>23 external speakers/attendees (3 UK, 20 international). Total funding £25,900.<br>Workshop website: <a href="https://icms.ac.uk/activities/workshop/mapping-class-groups/">icms.ac.uk/activities/workshop/mapping-class-groups/</a>                                                                                                                                                                                                                                                              | Sep 2025                                         |
|                                   | <b>Special Topology Seminar, University of Edinburgh</b><br>Lead organiser. To celebrate induction of Alan Reid as a Fellow of the Royal Society of Edinburgh.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | June 2024                                        |
|                                   | <b>Research Workshop: “Algebraic Methods in 4-Manifold Topology”</b><br>Organiser (with Mark Powell). 7 external speakers (2 UK, 5 international). Total funding £8,000.<br>Workshop website: <a href="https://maths.gla.ac.uk/~jnicholson/glasgow2023/">maths.gla.ac.uk/~jnicholson/glasgow2023/</a>                                                                                                                                                                                                                                                                                                                                        | June 2023                                        |
|                                   | <b>Heilbronn Show &amp; Tell Seminar</b><br>Organiser (with Alex Best and Caleb Springer)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2022 - 2023                                      |
| DEPARTMENTAL CONTRIBUTIONS        | <b>Imperial Undergraduate Research Prize</b> Judge/Reviewer<br><b>PhD Staff-Student Consultative Committee (PGRSSCC), UCL</b><br><b>Undergraduate Admissions Interviewer for UCL</b><br><b>Mathematics Faculty Board, Cambridge Student Representative</b>                                                                                                                                                                                                                                                                                                                                                                                   | 2022 - 2023<br>2019 - 2021<br>2018<br>2015, 2016 |

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| OUTREACH AND SERVICE TO EDUCATION         | <b>Talks at two London Schools</b> Attended by roughly 75 and 250 students respectively                                                                                                                                                                 | 2022                                                                             |
|                                           | <b>ROSS Mathematics Program</b> Based in Columbus, Ohio & Huangshan, China                                                                                                                                                                              |                                                                                  |
|                                           | <i>Lecturer in Ross USA</i> ( $2 \times 1$ week): Introduction to Algebraic Topology                                                                                                                                                                    | 2019, 2021                                                                       |
|                                           | <i>Counselor in Ross Asia</i> ( $2 \times 5$ weeks): supervised students, led seminars                                                                                                                                                                  | 2017, 2018                                                                       |
|                                           | <b>Chalkdust Magazine</b> <i>Member of Editorial Team</i>                                                                                                                                                                                               | 2019 - 2020                                                                      |
|                                           | <b>Advanced Mathematics Support Programme, University College London</b>                                                                                                                                                                                |                                                                                  |
|                                           | <i>Teacher:</i> STEP II and III course (Cambridge Entrance Exam)                                                                                                                                                                                        | 2018, 2019                                                                       |
| SELECTED RESEARCH VISITS ( $\geq 1$ WEEK) | <b>Co-author: Edexcel A-level Further Mathematics, Core Pure 2</b>                                                                                                                                                                                      | 2018                                                                             |
|                                           | Core curriculum textbook for high school students in the UK. Available from Amazon: <a href="https://amazon.co.uk/Edexcel-level-Further-Mathematics-Textbook/dp/1292183349/">amazon.co.uk/Edexcel-level-Further-Mathematics-Textbook/dp/1292183349/</a> |                                                                                  |
|                                           | <b>University of Cambridge</b>                                                                                                                                                                                                                          |                                                                                  |
|                                           | <i>Teacher and Mentor:</i> Millenium Mathematics Project (volunteering in schools), Sutton Trust Summer Schools (x2), STEP Easter School (preparing students for STEP exam)                                                                             | 2015 - 2017                                                                      |
|                                           | <i>CAMBassador:</i> Helped to run regular events for underprivileged students                                                                                                                                                                           | 2015 - 2017                                                                      |
|                                           | <b>Underground Mathematics</b> <i>Resource Developer</i> (6 weeks) Created resources for use in UK 16-18 education (see <a href="https://undergroundmathematics.org/">undergroundmathematics.org/</a> )                                                 | 2015                                                                             |
|                                           | Max Planck Institute for Mathematics (Visiting Grigori Avramidi, 2 weeks)                                                                                                                                                                               | Sep 2024                                                                         |
| RESEARCH VISITORS HOSTED ( $\geq 1$ WEEK) | University of Augsburg (Visiting Jim Davis, 1 week)                                                                                                                                                                                                     | July 2023                                                                        |
|                                           | University of Southampton (Visiting Daniel Kasprowski, 1 week)                                                                                                                                                                                          | May 2023                                                                         |
|                                           | Fields Institute (Visiting Ian Hambleton, 1 month)                                                                                                                                                                                                      | April 2023                                                                       |
|                                           | Max Planck Institute for Mathematics (Visiting Ian Hambleton, 1 week)                                                                                                                                                                                   | June 2022                                                                        |
|                                           | Durham University (Visiting Mark Powell, 2 weeks)                                                                                                                                                                                                       | Feb, May 2022                                                                    |
|                                           | Indiana University Bloomington (Visiting Jim Davis, 3 weeks)                                                                                                                                                                                            | Apr 2022                                                                         |
|                                           | Max Planck Institute for Mathematics (Visiting Jim Davis, 2 months)                                                                                                                                                                                     | Mar-Apr 2021                                                                     |
| RESEARCH VISITORS HOSTED ( $\geq 1$ WEEK) | Tommy Hofmann, 1 week, funded by G&T seminar and University of Siegen                                                                                                                                                                                   | Nov 2025                                                                         |
|                                           | Ian Hambleton, 2 weeks, funded by Heilbronn Small Grant                                                                                                                                                                                                 | Oct 2025                                                                         |
|                                           | Grigori Avramidi, 1 week, funded by G&T seminar and Max Planck Institute                                                                                                                                                                                | Jan 2025                                                                         |
|                                           | Ian Hambleton, 1 week, funded by Heilbronn Small Grant                                                                                                                                                                                                  | Nov 2024                                                                         |
|                                           | Jim Davis, 2 weeks, funded by Heilbronn Small Grant                                                                                                                                                                                                     | June 2024                                                                        |
|                                           | Ian Hambleton, 2 weeks, funded by Isaac Newton Institute                                                                                                                                                                                                | May 2024                                                                         |
|                                           | Simona Veselá, 1 week, funded by John Robertson Bequest                                                                                                                                                                                                 | Apr 2024                                                                         |
| SKILLS                                    | Daniel Kasprowski, 1 week, funded by John Robertson Bequest                                                                                                                                                                                             | Apr 2024                                                                         |
|                                           | <b>Programming and markup languages</b>                                                                                                                                                                                                                 |                                                                                  |
|                                           | Magma, GAP, L <sup>A</sup> T <sub>E</sub> X (advanced)                                                                                                                                                                                                  |                                                                                  |
| REFERENCES                                | SageMath, Python, MATLAB, Mathematica, HTML, Bash (proficient)                                                                                                                                                                                          |                                                                                  |
|                                           | <b>Languages</b> English (native), Spanish (advanced)                                                                                                                                                                                                   |                                                                                  |
|                                           | Mark Powell ( <i>Professor, University of Glasgow</i> )                                                                                                                                                                                                 | <a href="mailto:mark.powell@glasgow.ac.uk">mark.powell@glasgow.ac.uk</a>         |
|                                           | Martin Bridson ( <i>Professor, University of Oxford</i> )                                                                                                                                                                                               | <a href="mailto:martin.bridson@maths.ox.ac.uk">martin.bridson@maths.ox.ac.uk</a> |
|                                           | Ian Hambleton ( <i>Professor, McMaster University</i> )                                                                                                                                                                                                 | <a href="mailto:hambleton@mcmaster.ca">hambleton@mcmaster.ca</a>                 |